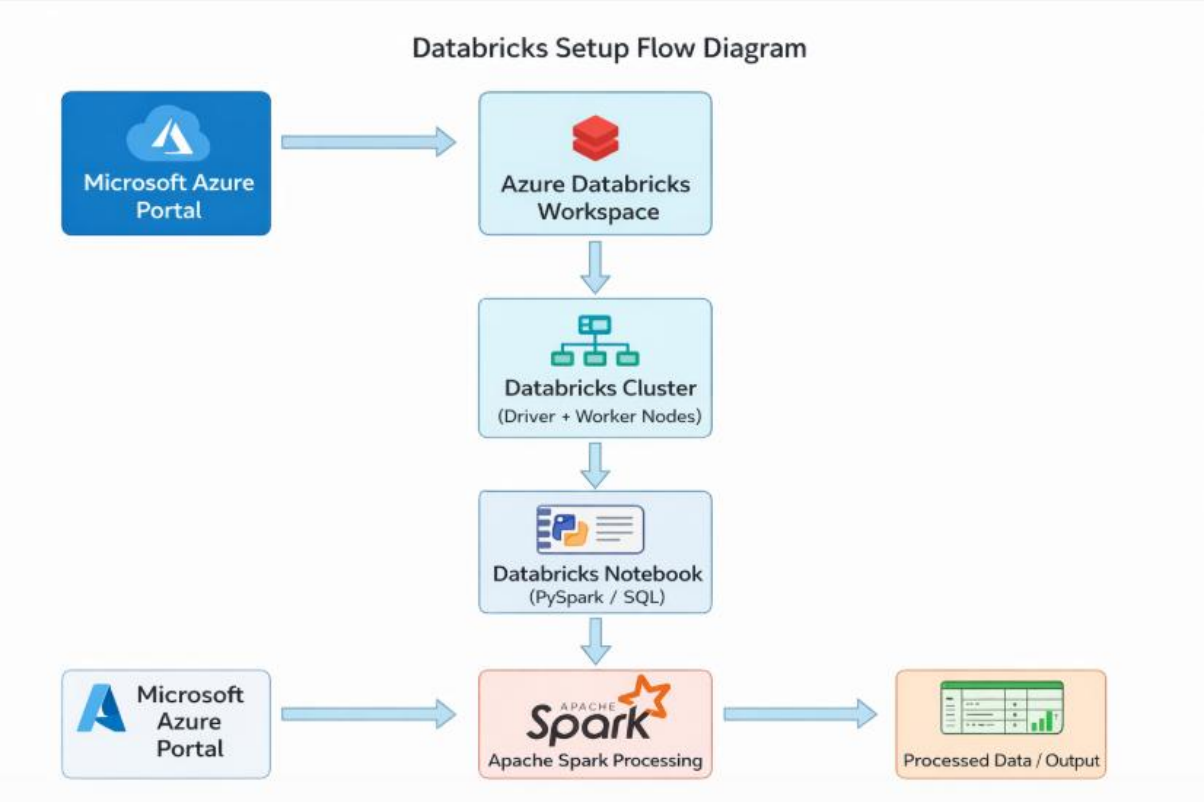
Databricks setup

Objective of Azure Databricks Setup

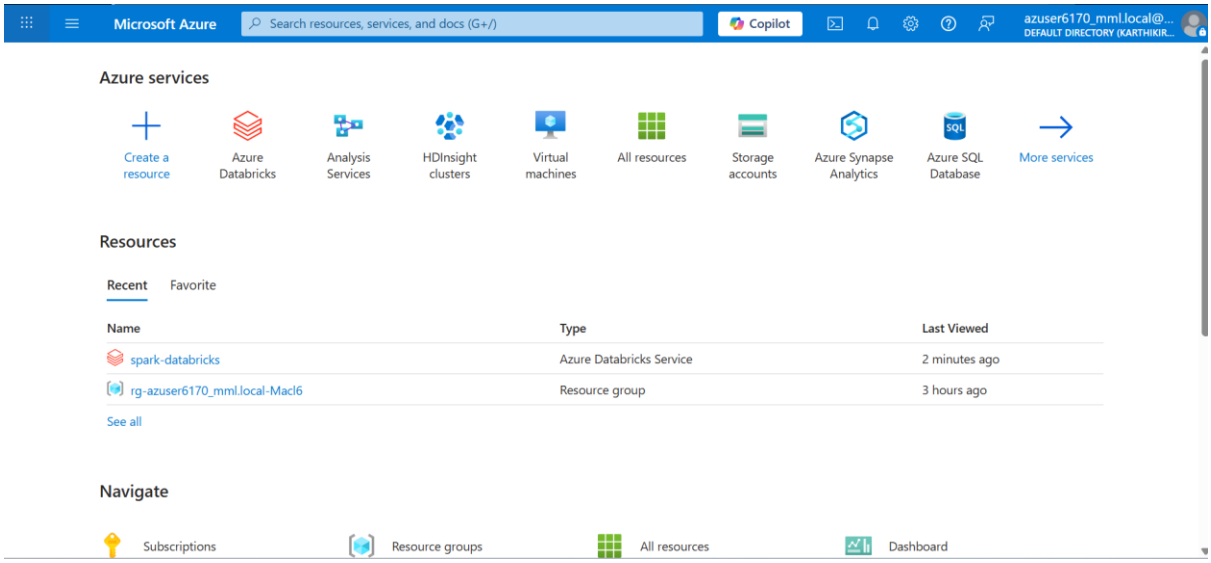
The objective of this setup is to create a cloud-based big data processing environment using Azure Databricks. It involves provisioning an Azure Databricks workspace and configuring a Spark cluster to enable scalable and distributed data processing. This setup allows users to execute PySpark code through notebooks, validate cluster functionality by running sample commands, and perform data engineering and ETL operations efficiently.

Databricks Setup Flow Diagram



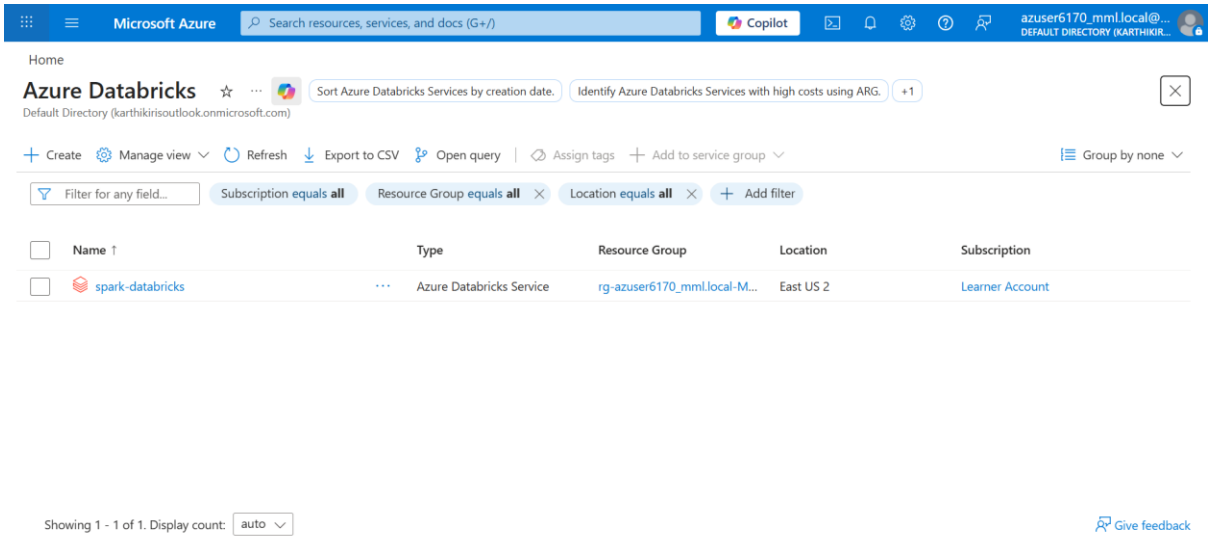
Step 1: Azure Portal Home

- Login to Microsoft Azure Portal
- Navigate to **Azure Services**
- Click on **Azure Databricks**



Step 2: Databricks Resource List

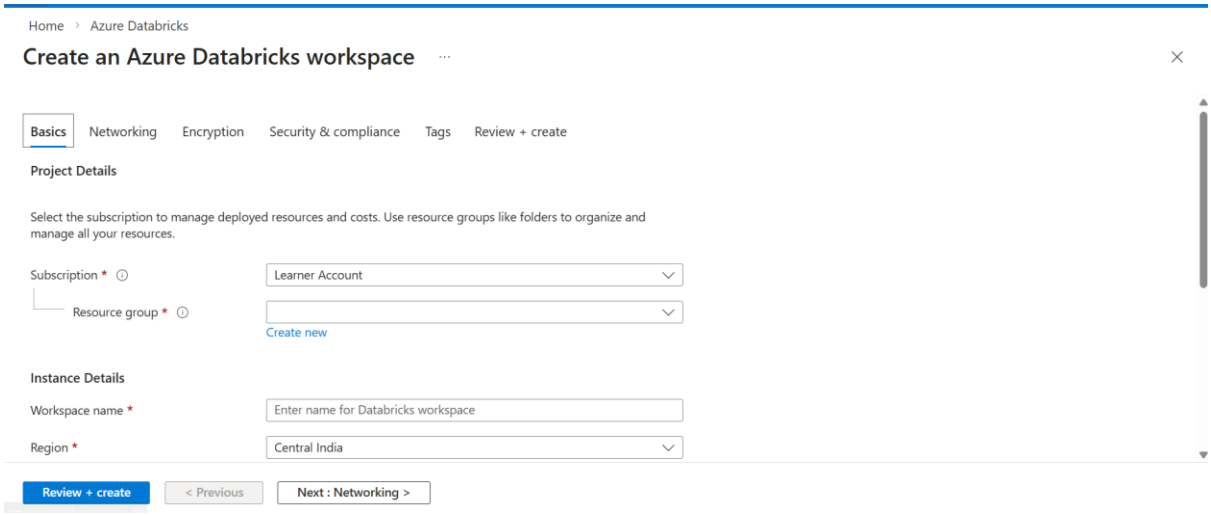
- View existing Databricks resources
- Click on your workspace (e.g., spark-databricks)
- Verify Resource Group and Subscription



Step 3: Create Databricks Workspace – Basics

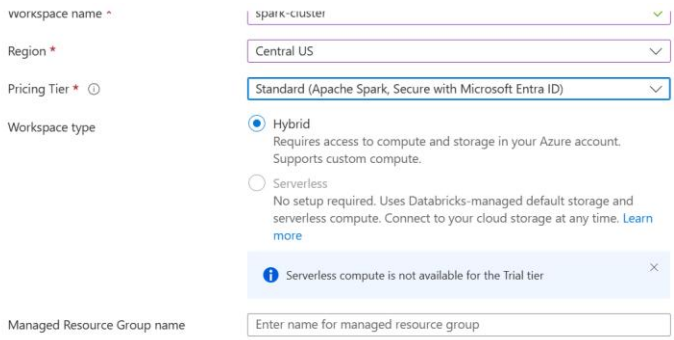
- Click **Create Resource → Azure Databricks**
- Fill the following:
  - Subscription: Learner Account
  - Resource Group: Select existing / Create new

- Workspace Name: Enter name



Step 4: Fill Workspace Details

- Provide required details:
  - Workspace Name: spark-cluster
  - Region: Central US
  - Pricing Tier: Standard
  - Workspace Type: Hybrid

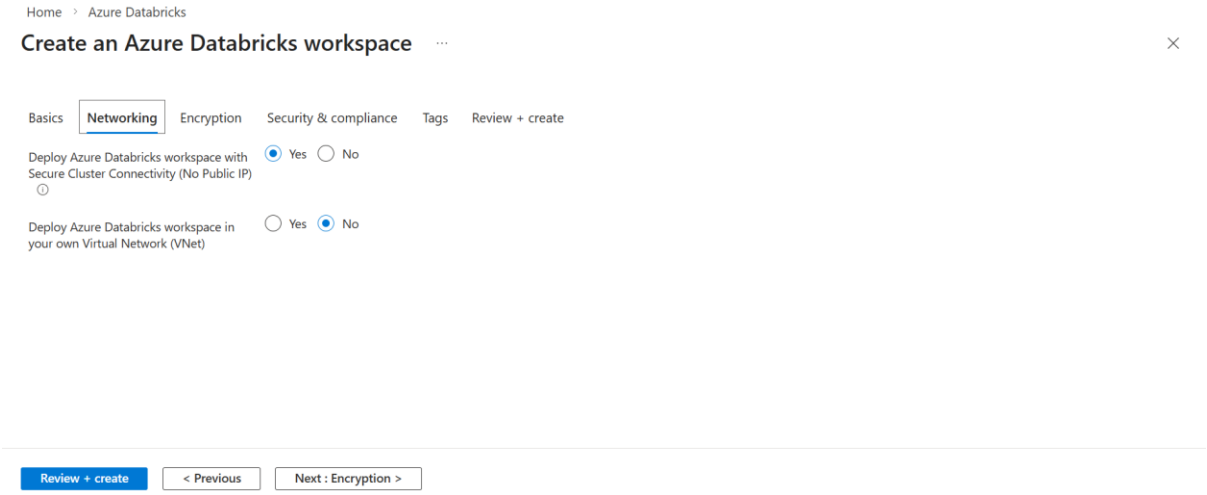


Step 5: Networking Configuration

- Configure networking settings:
  - Secure Cluster Connectivity: Yes

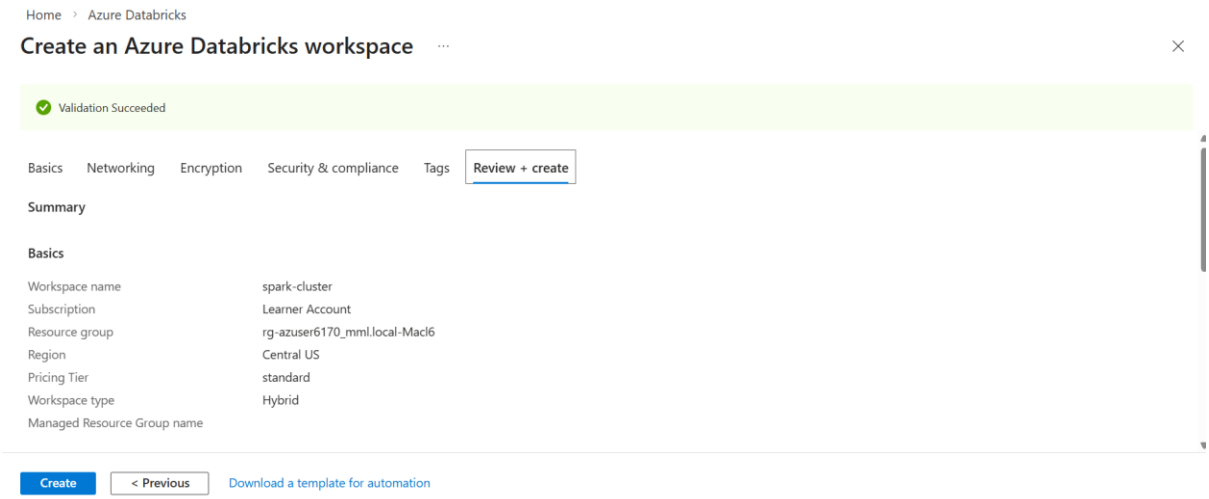
- Virtual Network: No (default)

- Click **Next**



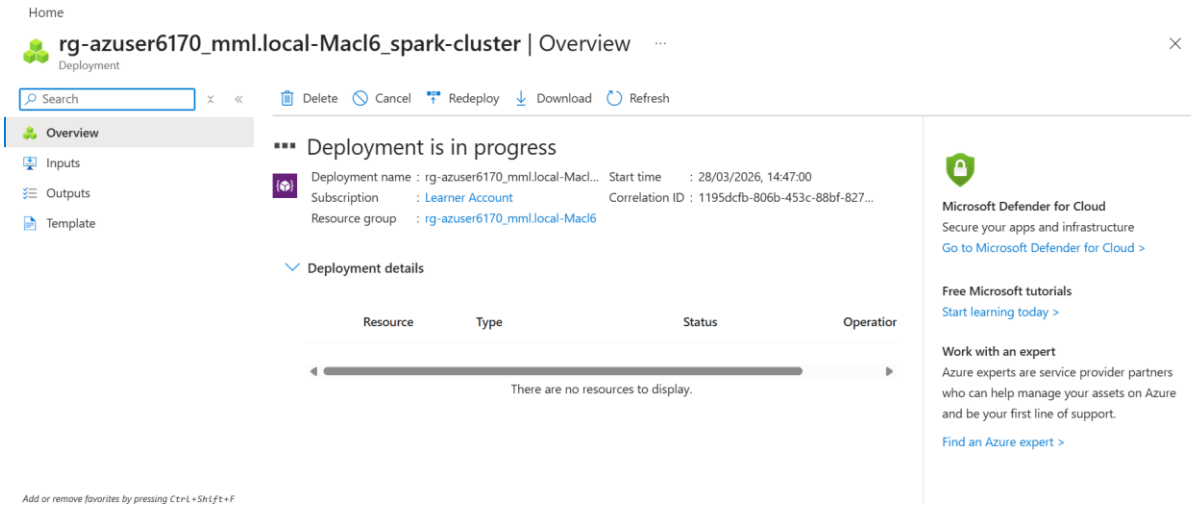
Step 6: Review and Create

- Review all configurations
- Validate settings
- Click **Create**



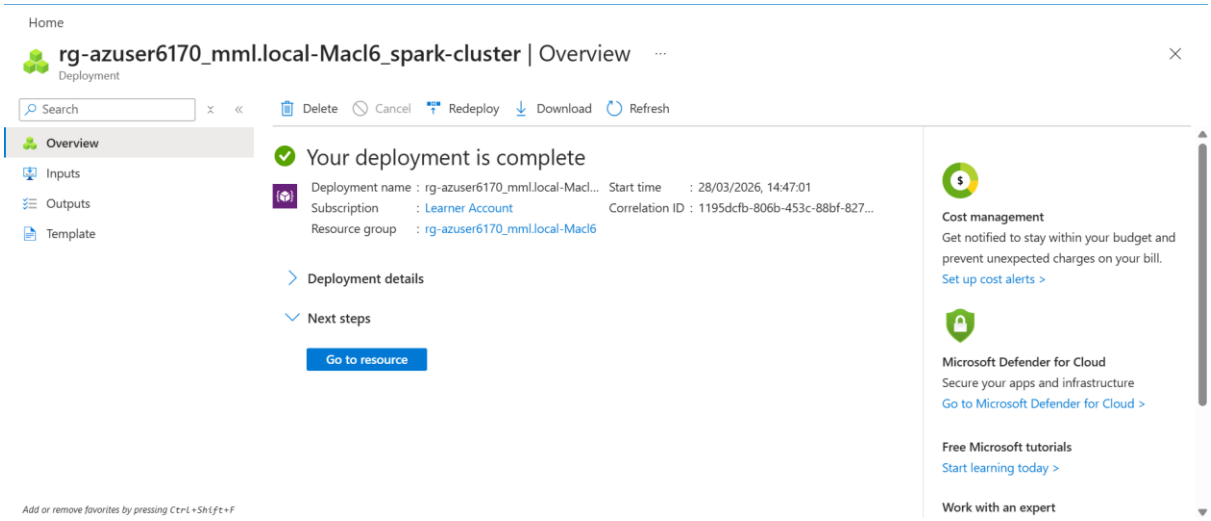
Step 7: Deployment in Progress

- Azure starts deployment
- Status shows: **Deployment is in progress**
- Wait for completion



Step 8: Deployment Completed

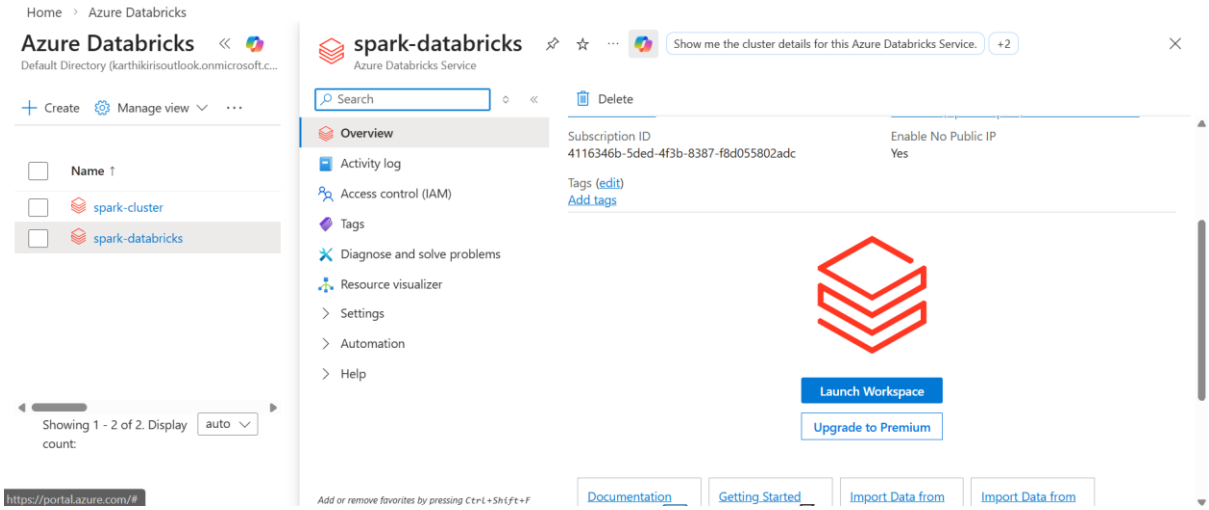
- Deployment successful
- Click **Go to Resource**



Step 9: Databricks Workspace Home

- Databricks workspace is launched
- You can see:
    - Workspace menu
    - Compute
    - Catalog

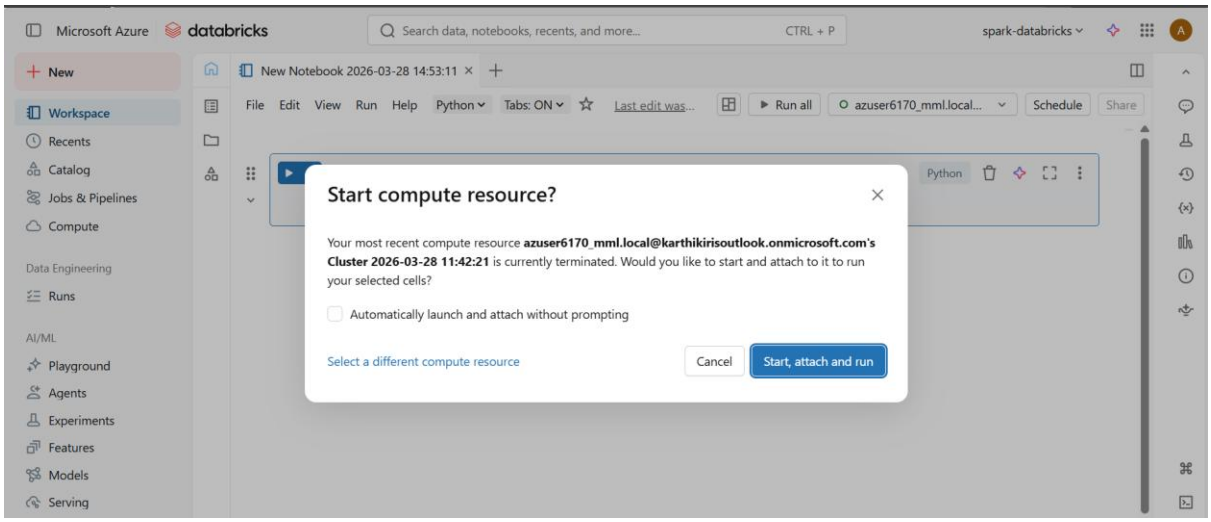
- Jobs



Step 10: Create Notebook in Databricks

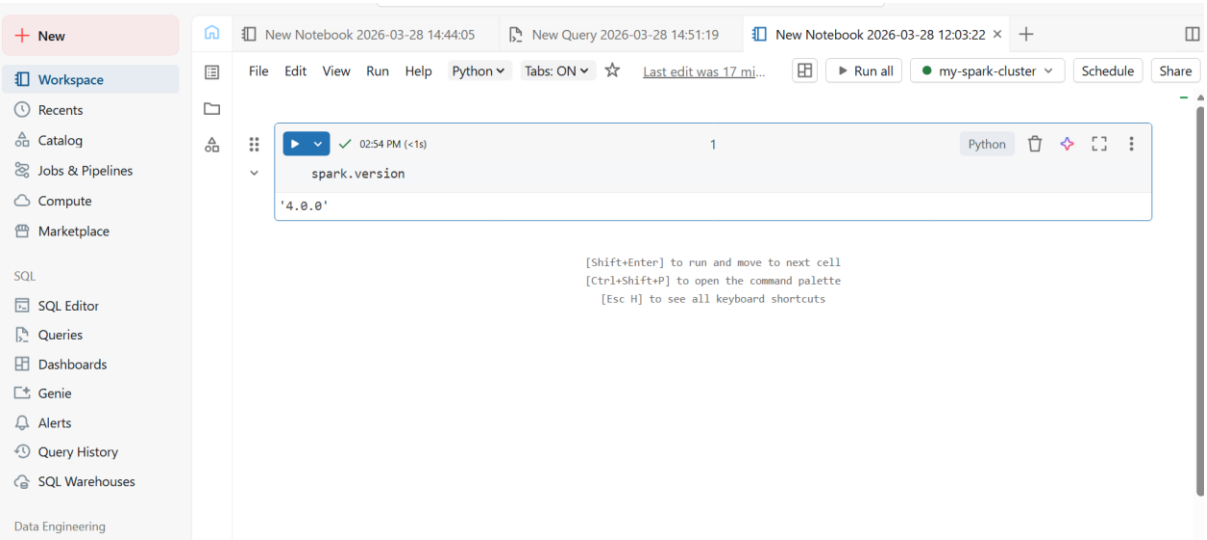
After opening Databricks workspace, click on **New**

- From dropdown, select **Notebook**
- This option is used to create a new notebook where we write PySpark code



Step 11: Run Spark Code

- After cluster starts, notebook is connected
- Run the code:

spark.version

- Output displays Spark version (e.g., **4.0.0**)
- This confirms Spark cluster is working successfully